Deep Dive 03 | Alerts and CAP Processing

Trace alert retrieval, summary normalization, CAP XML parsing and ticker behavior.

React/Vite Edition | Repository-grounded study material | Snapshot: 2026-10-24

1. Server summary route

backend/routes/alertRoutes.js posts to the NDMA SACHET FetchAllAlertDetails endpoint. Each source row is mapped to a common object containing kind, stable uid, identifier, title/event, time, location, severity, detail link, XML link and optional coordinates. Results are sorted by time and limited to five. Cache-Control permits short browser caching and stale-while-revalidate.

2. CAP detail parsing

GET /api/alerts/cap/:identifier removes an IN prefix when constructing the source XML request. fast-xml-parser converts the response into an object. Helper text() handles strings, numbers and common text-node objects. The route selects the English info block when present, normalizes area entries and returns headline, severity, urgency, certainty, timestamps, description, instructions and areas.

3. Frontend rendering

- AlertTicker fetches /api/alerts on mount and every minute.
- It loads readable CAP details for rows with identifiers and uses the human-friendly detail when available.
- Notification components share alert details through frontend/src/lib/alertDetails.js.
- CapAlertPage renders the details and links back to the original CAP XML/SACHET dashboard.

4. Other feed utilities

frontend/src/lib/usgs.js parses a USGS GeoJSON feed and maps magnitude, place, time and coordinates to earthquake alert objects. frontend/src/lib/gdacs.js parses a GDACS RSS feed and extracts flood event metadata. These utilities require VITE_* feed URLs. Confirm call sites before describing them as part of the active ticker; repository utilities can exist without being used in the current visible path.

5. Failure and correctness concerns

- SACHET request errors return an empty items array and an error field with HTTP 200; UI may confuse provider failure with no active alerts.
- Invalid or non-array provider response should be validated before rows.map.
- CAP identifiers are normalized differently between summary and detail route; test actual identifier forms.
- Alert timestamps may be missing or invalid; sort must remain stable for null times.
- CAP XML can contain repeated elements and different languages; parser settings and array normalization matter.
- Alert content is external input and must be escaped in the UI.

6. Improvements and exercises

- Return explicit provider status or a freshness timestamp and distinguish empty from unavailable.
- Add timeout and safe schema validation for summary and detail routes.
- Cache last successful normalized response and make its age visible.
- Mock one alert with multiple languages and multiple areas.

- Test missing identifier, malformed XML, provider timeout and unknown alert ID.

Safety framing: an alert display is a convenience view of source data, not a guaranteed warning channel. Users should follow official alerts and local instructions.